

Project Overview: iOS & macOS CRUD and Chat

Task 1: iOS CRUD App

- Develop an iOS app using SwiftUI to perform CRUD operations against a PostgreSQL database.
- Store any data you choose (e.g., notes, tasks, inventory).
- Use URLSession for REST calls to the backend.

Requirements:

- Backend: Python (FastAPI or Django)
- Database: PostgreSQL in Docker

Task 2: iOS Chat Tab

- Add a Chat tab within the iOS app to select between two modes:
 - - Chat with another user (yourself on macOS).
 - - Chat with ChatGPT or an LLM agent (future step).
- Implement UI to switch between conversation targets.

Requirements:

- Backend: Python (FastAPI or Django)
- Real-time: WebSockets via FastAPI/Django Channels

Task 3: macOS Chat App

- Create a macOS app target in the same Xcode workspace for chat.
- Connect to the same WebSocket endpoint as the iOS app.
- Enable two-way chat between iOS and macOS clients.
- Persist chat history in MongoDB.

Requirements:

- Database: MongoDB in Docker
- Backend: Python (FastAPI or Django) with MongoDB driver

Task 4: OpenAPI/LLM Integration

- Extend the chat backend to call the OpenAI API (or another LLM endpoint).
- Detect when the user selects ChatGPT mode and forward messages to the LLM.
- Return LLM-generated responses over WebSocket to clients.

Requirements:

- Backend: Python using the OpenAI Python SDK
- API Key managed via environment variables

